X	f	fX	$\|D\| = \|X - 8\|$	f|D|
2	2	4	6	12
4	2	8	4	8
6	4	24	2	8
8	5	40	0	0
10	3	30	2	6
12	2	24	4	8
14	1	14	6	6
16	1	16	8	8
	$N = 20$	$\Sigma fX = 160$		$\Sigma f \mid D \mid = 56$

Table : 8.15

Example 8.22

Calculate the Mean deviation about median and its relative measure for the following data.

X	15	25	35	45	55	65	75	85
frequency	12	11	10	15	22	13	18	19

Solution:

Already the values are arranged in ascending order then Median is obtained by the following:

X	f	cf
15	12	12
25	11	23
35	10	33
45	15	48
55	22	70
65	13	83
75	18	101
85	19	120
	N = 120	

Table : 8.16

$$\text{Median} = \text{size of } \left(\frac{(n+1)}{2}\right)^{th} \text{ value}$$

$$= \text{size of } \left(\frac{(120+1)}{2}\right)^{th} \text{ value}$$

$$= \text{size of } 60.5^{th} \text{ item} = 55$$

MD about Median

$$= \frac{\sum f|X - Median|}{N} = \frac{\sum f|D|}{N}$$

Mean deviation about Median

X	f	$\|D\| = \|X - 55\|$	f|D|
15	12	40	480
25	11	30	330
35	10	20	200
45	15	10	150
55	22	0	0
65	13	10	130
75	18	20	360
85	19	30	570
	N = 120		$\Sigma f \mid D \mid = 2220$

Table : 8.17

MD about Median $= \frac{2220}{120} = 18.5$

Coefficient of mean deviation about median

$$= \frac{MD \text{ about } Median}{Median}$$

$$= \frac{18.5}{55} = 0.34$$

Example 8.23

Find out the coefficient of mean deviation about median in the following series

Age in years	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of persons	20	25	32	40	42	35	10	8

Calculations have to be made correct to two places of decimals.

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